Diagonalization

Transformation

↓

$Ax = \lambda x$

Eigen vector for

$A^2 x \longrightarrow A\frac{Ax}{\lambda} = A\lambda x$

$= \lambda Ax = \lambda^2 x$

$\therefore A^2 x = \lambda^2 x$

Scaled by λ^2

n times A applied $\rightarrow \lambda^n$ times scaled.

$A^{-1}x$

$A^{-1}Ax = A^{-1}\lambda x$

$= \lambda A^{-1}x$

$\Rightarrow x = \lambda$

$\therefore A^{-1}x = \frac{1}{\lambda}x$

λ → can't be zero

if A^{-1} exists the x can't be zero.

When does inverse exist?

if all the col has pivots.

Nullspace of an invertible Mat.

→ It is only 0 vector.

$Ax = 0$

↓

can't be anything rexthere than 0.

So x is a nonzero vector.

Only 0 vect gets scaled by 0.

λ can't be 0 as A^{-1} exists.

#

$A \longrightarrow n$ independent eigen vectors

$n \times n$

$$S = \begin{bmatrix} | & | & & | \\ x_1 & x_2 & \cdots & x_n \\ | & | & & | \end{bmatrix}$$

matrix of eigenvectors.

cols are dependent?

$$\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \begin{bmatrix} | & | & & | \\ x_1 & x_2 & \cdots & x_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ \lambda_1 x_1 & \lambda_2 x_2 & \cdots & \lambda_n x_n \\ | & | & & | \end{bmatrix}$$

A S

$A x_1 = \lambda_1 x_1$

↓

eigenvector of A

$$= \begin{bmatrix} | & | & & | \\ x_1 & x_2 & \cdots & x_n \\ | & | & & | \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & 0 & 0 \\ 0 & \lambda_2 & 0 & 0 \\ 0 & 0 & \lambda_3 & \\ \vdots & \vdots & & \ddots \\ & & & \lambda_n \end{bmatrix}$$

S

Λ

capital lambda.

$$AS = S\Lambda$$

$$\boxed{A = S\Lambda S^{-1}}$$

→ very important!

hinges on the idea S^{-1} exists.

When does it not?

Matrix was not diagonalizable

As S didn't have independent col.

Find a Mat where

The $\lambda_1, \lambda_2 \ldots$ are same

but x_1, x_2 are different.

↳ Identity Matrix.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^2 = (S\Lambda S^{-1})(S\Lambda S^{-1})$$

(cancel)

$$= S\Lambda^2 S^{-1}$$

Eigen vector same

" value squared

for A^k,

$$S\Lambda^k S^{-1}$$

I like see
A series that follows a pattern

$u_1 = Au_0$ | $u_2 = Au_1 = A^2u_0$

$u_k = A^k u_0$

Fibonacci series. → need O(1)

$F_{k+2} = F_{k+1} + F_k$ → O(n)

Need a system of eqn to solve with LA so insert dummy eqn.

$$F_{k+1} = F_{k+1}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix} = \begin{bmatrix} F_{k+2} \\ F_{k+1} \end{bmatrix}$$

A $\quad$ x $\quad$ b

$AF_k = F_{k+1}$

↓

Encodes the relationship of the Fibonacci

$F_1 = AF_0$

$F_2 = A^2F_0$

$F_{1000} = A^{1000}F_0$

series

$$\begin{vmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = 0$$

$(A - \lambda I)$

$$-\lambda + \lambda^2 - 1 = 0$$

$$\lambda^2 - \lambda - 1 = 0$$

$$\lambda = \frac{1 \pm \sqrt{5}}{2}$$

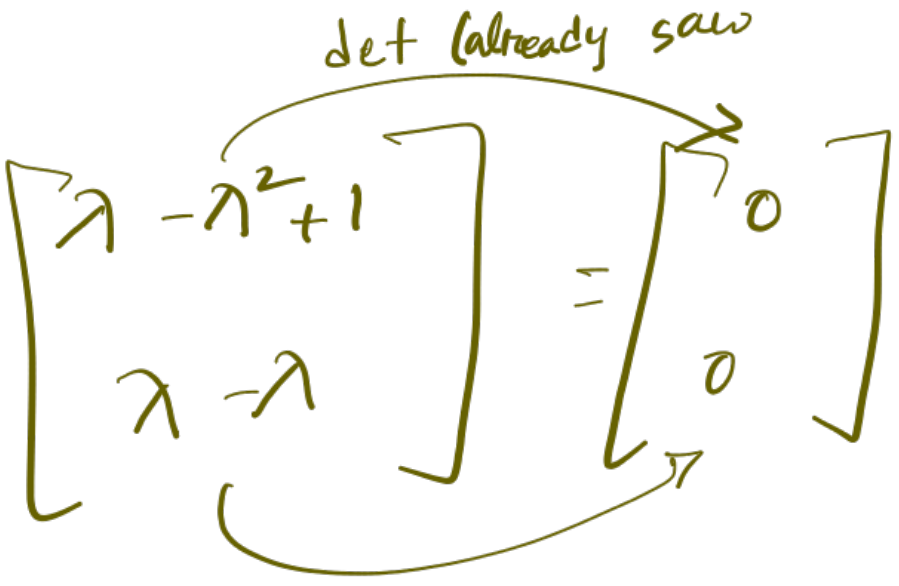
$$\lambda_1 = \frac{1+\sqrt{5}}{2} \qquad \lambda_2 = \frac{1-\sqrt{5}}{2}$$

Golden Ratio → rate of increase of Fibonacci series.

got E.vals. Need E.vects.

$$\begin{bmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{bmatrix} \begin{bmatrix} x \\ \end{bmatrix} = 0$$

$$\hookrightarrow \begin{bmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{bmatrix} \begin{bmatrix} \lambda \\ 1 \end{bmatrix} = 0$$



So, $x_1 = \begin{bmatrix} \lambda_1 \\ 1 \end{bmatrix}$, $x_2 = \begin{bmatrix} \lambda_2 \\ 1 \end{bmatrix}$

$$S = \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \quad \star$$

$$S^{-1} = \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{bmatrix}$$

$$A^k u_0 = S \Lambda^k S^{-1} u_0$$

$$= \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{bmatrix} \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$=$$

$$\begin{bmatrix} F_{k+2} \\ F_{k+1} \end{bmatrix} \to 0$$

$$1 + 0$$

$$\begin{bmatrix} F_{k+1} \\ \end{bmatrix}$$

$-1+0$

$\rightarrow 1$

$$= \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{bmatrix} \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1^k \\ -\lambda_2^k \end{bmatrix}$$

$$= \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} \lambda_1^{k+1} - \lambda_2^{k+1} \\ \lambda_1^k - \lambda_2^k \end{bmatrix}$$

1000th term

k^{th} term

A Vector!

1001th term

Not Mat.

$$F_{1000} = \frac{\lambda_1^{1000} - \lambda_2^{1000}}{\lambda_1 - \lambda_2}$$

O(1) Soln to find the 1000th term of Fibonacci series.

Geometric series

↳ 2nd term is multiplication of the 1st term.

↓ this works for any geometric

series.